

Accessibility by Design:

Practical A11y for Phoenix & LiveView

That's me

- Software Engineer
 - Elixir/Phoenix,
 - Postgres,
 - HTML, CSS, JS
- Conferences
- Meetup Organiser
- Team Lead, Tech Lead and Project Lead



... also me

- Family Friends, the pack:
 - 5 cats
 - 2 dogs
 - 2 pigs
- Lives in the middle of nowhere in Franconia (Bavaria)



*The goal is not to make everyone an accessibility expert,
but to make accessibility a natural part
of our everyday engineering work.*

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01

The What

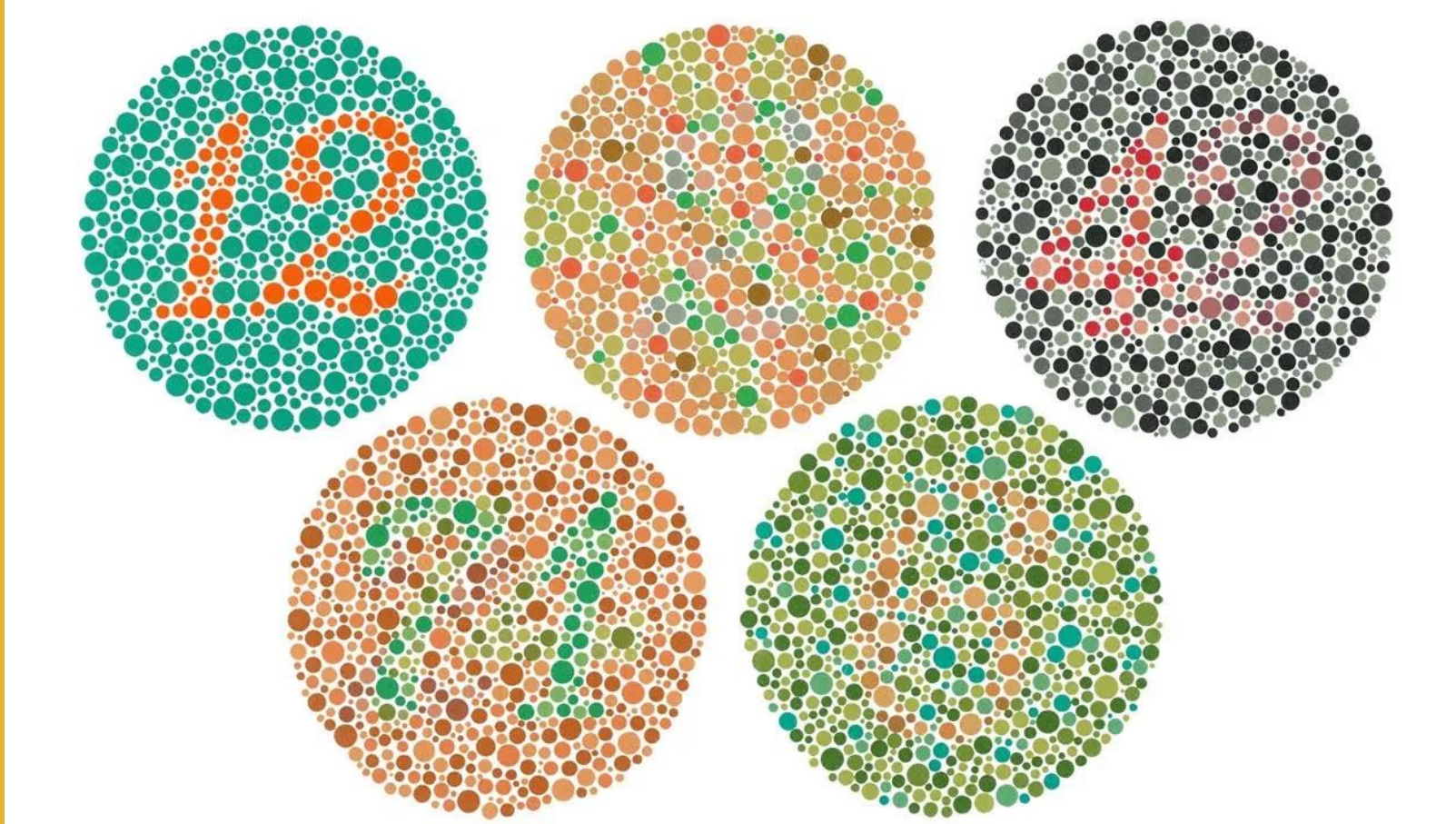
Different (views on) disabilities

Who of you is disabled in some way?

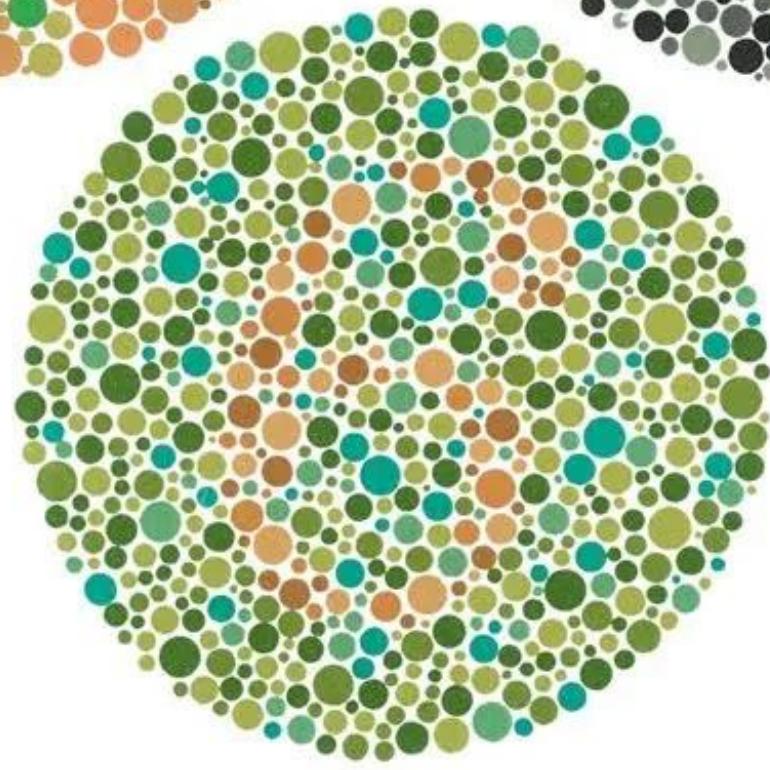
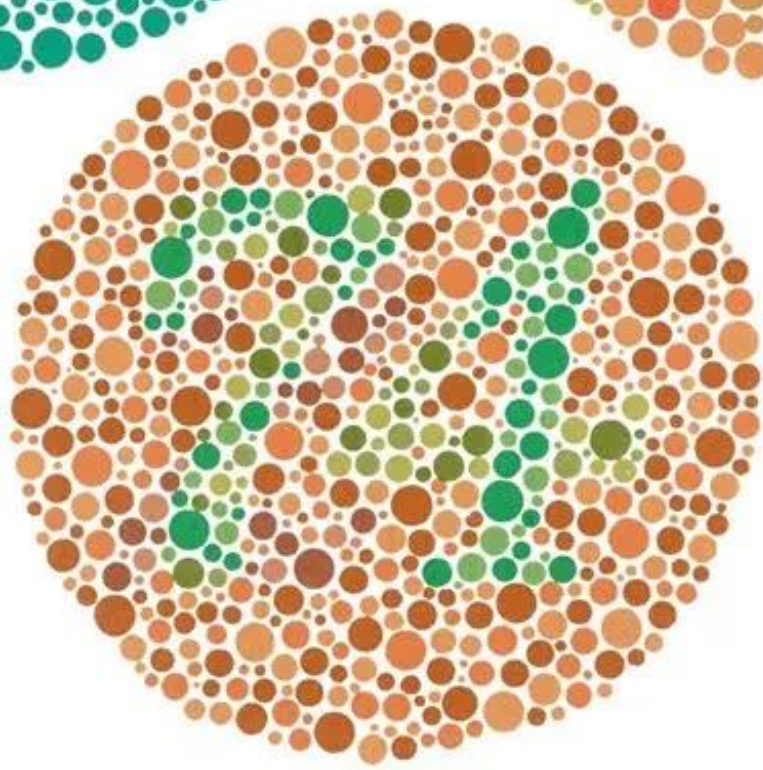
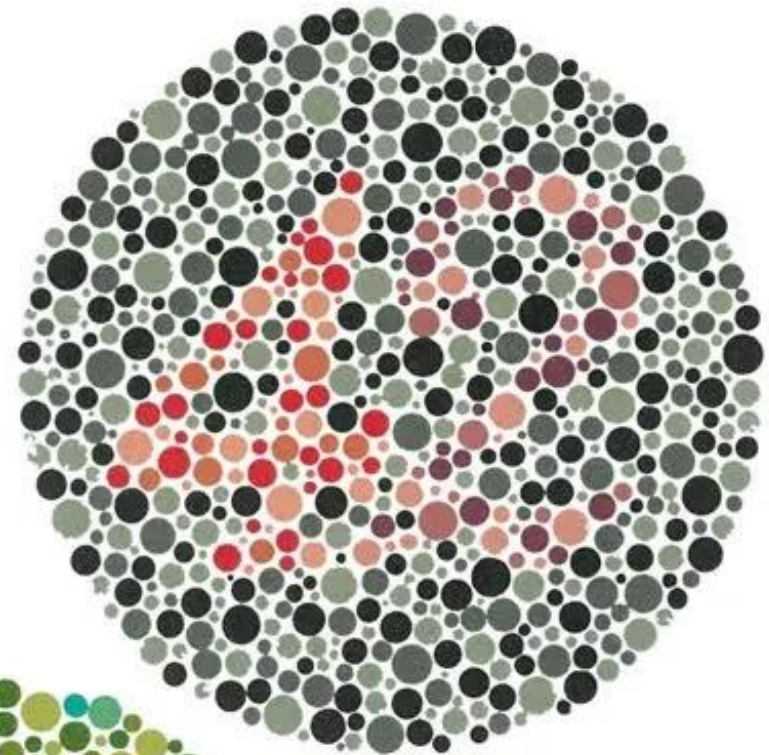
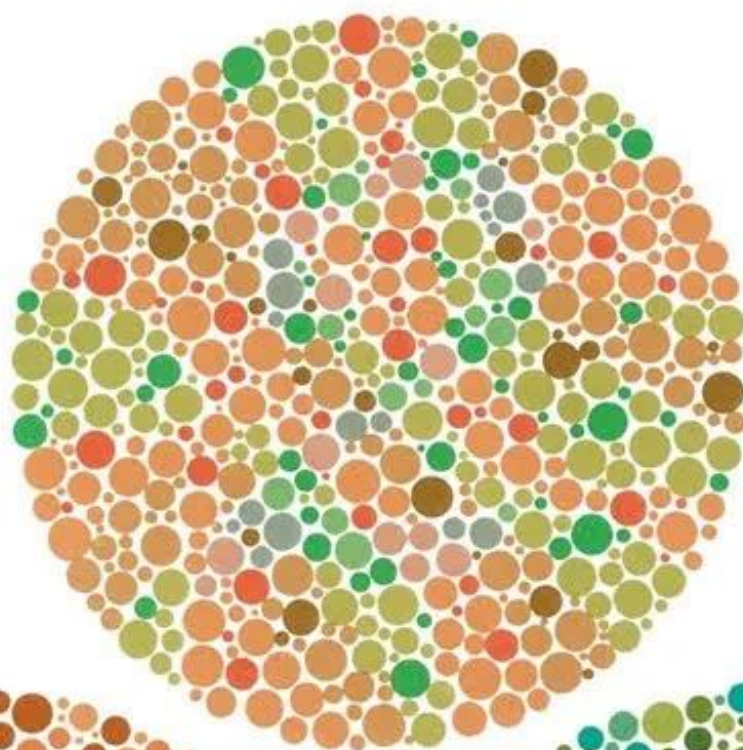
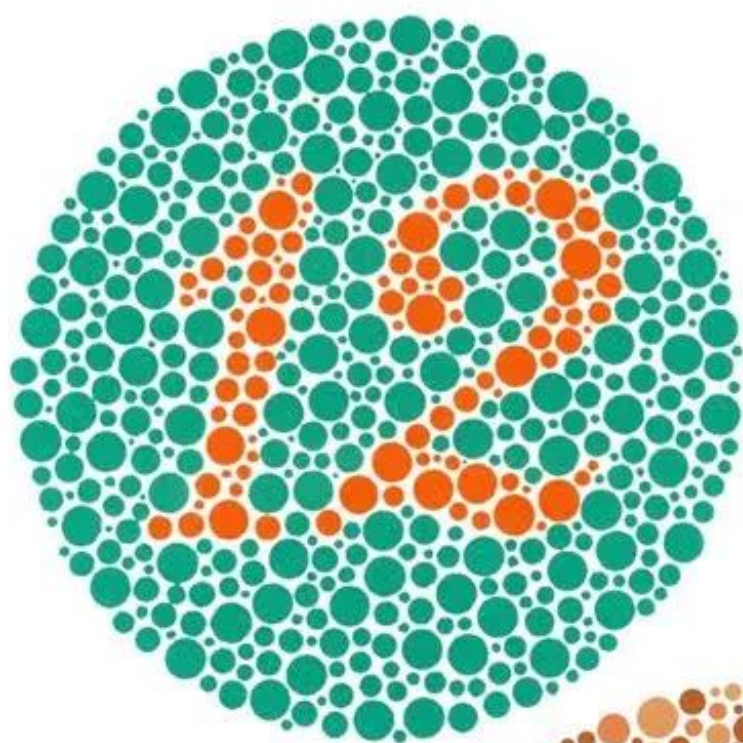
Different (views on) disabilities

	Permanent	Temporary	Situational
 Touch	<i>One arm</i>	<i>Arm injury</i>	<i>Carry stuff</i>
 See	<i>Blind</i>	<i>Cataract</i>	<i>Bright sunlight</i>
 Hear	<i>Deaf</i>	<i>Ear infection</i>	<i>Loud environment</i>
 Speak	<i>Non-verbal</i>	<i>Laryngitis</i>	<i>Heavy accent</i>
 Think	<i>Cognitive disability</i>	<i>Fatigue</i>	<i>Distracted by multi-tasking</i>

Can you read the numbers ... easily?



Source: Derek Torsani, <https://medium.com/@dmtors/being-a-color-blind-designer-aa6828c09a42>



Have you ever ...

*... tried using your phone one-handed
while holding a baby?*

Have you ever ...

... struggled to read a screen because of bright sunlight?

Have you ever ...

... used subtitles on a video?

Who of you is disabled in some way?

Different (views on) disabilities: Web Development

	Permanent	Temporary	Situational
 Touch	<i>One arm</i>	<i>Arm injury</i>	<i>Carry stuff</i>
 See	<i>Blind</i>	<i>Cataract</i>	<i>Bright sunlight</i>
 Hear	<i>Deaf</i>	<i>Ear infection</i>	<i>Loud environment</i>
 Connect	<i>Low bandwidth</i>	<i>Data cap reached</i>	<i>On a train/tunnel</i>



LiveView already has accessibility and resilience by design!

A Fourth Kind of Disability: Your Network connection

This disability is situational for many, but structural for some.

A typical SPA

- ✗ Blank page → JS bundle → hydrate
- ✗ Step 2 can just ... not finish.

Phoenix LiveView

- ✓ Full HTML on the first request
- ✓ Usable before the WebSocket connects.

02

The Why

Empathy • Common Sense • European Accessibility Act

For whom is that feature useful?



... all of us 😊

Source: Nick-Philly, CC BY-SA 4.0, <https://commons.wikimedia.org/w/index.php?curid=84618901>

Why are we dealing with A11y?

01

For Others

To include the largest number of people possible to experience our app.

02

For Us

... that includes all of us.

03

For Compliance

To satisfy the European Accessibility Act.

European Accessibility Act — EAA

- When
 - In force since 28 June 2025 for new products and services
 - Legacy products must comply by 28 June 2030
- Who
 - Consumer-facing (B2C) products and services **sold** in the EU
- What
 - EU standard EN 301 549, which in practice means **WCAG 2.1 AA**

Web Content Accessibility Guidelines — WCAG

- International standard for web accessibility by the World Wide Web Consortium (W3C).

POUR Principles

PERCEIVABLE

P

Users must be able to see or hear the content

Does our icon only `<button>` ship with an accessible label?

OPERABLE

O

Users must be able to interact with the interface

phx-click on a `<div>`: fine with a mouse, useless by keyboard

UNDERSTANDABLE

U

Content and operation must be understandable

CoreComponents give you consistency for free

ROBUST

R

Content must be interpreted by assistive technologies

HEEx compiles to real HTML, not a virtual DOM

Web Content Accessibility Guidelines— WCAG

- International standard for web accessibility by the W3C.



- WCAG A, AA, AAA structured checklist.

03

The How

Assistive Tech • Accessibility Tree (LiveView edition) • Native Elements/ARIA • Testing • Pull Request Checklist

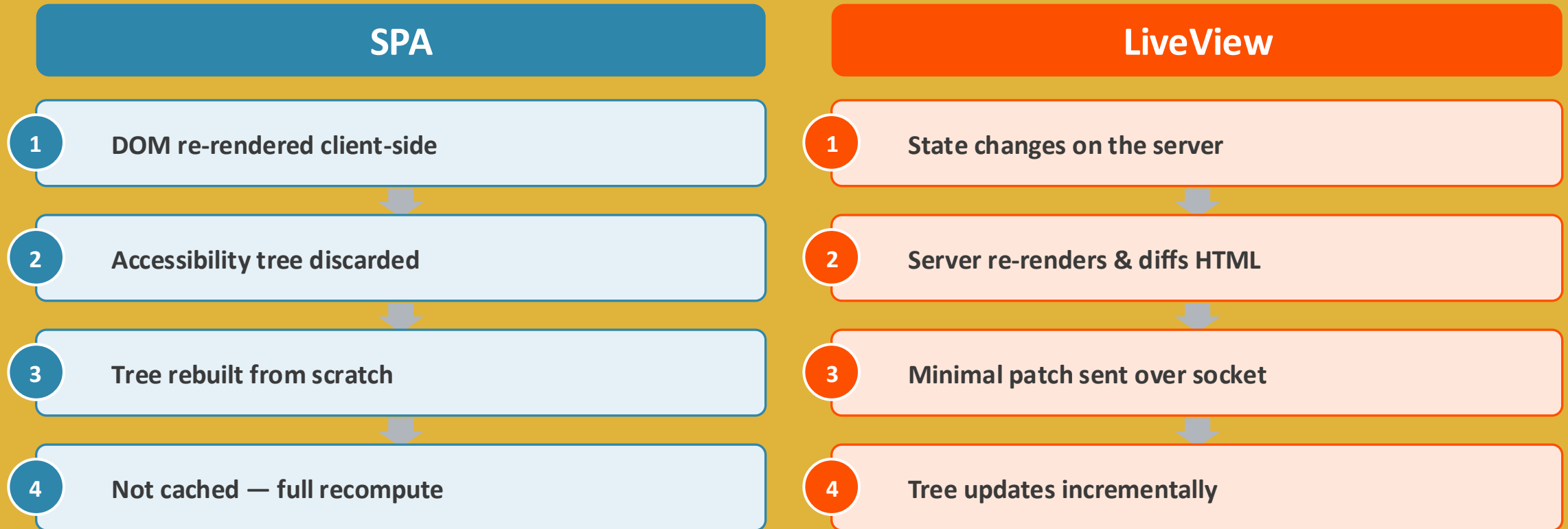
Assistive Tech for the Web

- Computer Access Tools
- Visual Aids and Magnification Systems
- Screen Readers
 - Text-to-Speech
 - They identify form fields
 - Interpret image alternative text
 - They navigate linearly or jump between elements like *headings, links, and landmarks*

HTML vs. DOM Tree vs. Accessibility Tree

- Let's take a look what the browser sees ...

The Accessibility Tree Cycle: SPA vs. LiveView



You have to announce some changes yourself

- Bad News: The browser won't always treat a patch as something new to announce.
- Classic case: `phx-update="stream"`
 - inserts a new chat message.
 - it shows instantly — but a screen reader gets nothing without an ARIA live region (`aria-live="polite"/"assertive"`).



Rule of thumb

If a sighted user notices a change, tell the screen reader, too.

The Web: HTML, CSS, JS

- HTML as nouns
gives meaning/structure
- CSS as adjectives
describes & decorates
- JS as verbs
adds behaviour

Most meaning and thus accessibility
wins
are in the noun box 📦



```
1 <div>
2   <span>Three words</span>
3   <div>
4     <a>one word</a>
5     <a>one word</a>
6     <a>one word</a>
7     <a>one word</a>
8   </div>
9 </div>
10 <div>
11   <div>
12     <div>five words</div>
13   </div>
14   <div>
15     <div>three words</div>
16     <div>forty-six words</div>
17     <div>forty-four words</div>
18   </div>
19   <div>
20     <div>seven words</div>
21     <div>sixty-eight words</div>
22     <div>forty-four words</div>
23   </div>
24 </div>
25 <div>
26   <span>five words</span>
27 </div>
```

```
1 <header>
2   <h1>Three words</h1>
3   <nav>
4     <a>one word</a>
5     <a>one word</a>
6     <a>one word</a>
7     <a>one word</a>
8   </nav>
9 </header>
10 <main>
11   <header>
12     <h2>five words</h2>
13   </header>
14   <section>
15     <h2>three words</h2>
16     <p>forty-six words</p>
17     <p>forty-four words</p>
18   </section>
19   <section>
20     <h3>seven words</h3>
21     <p>sixty-eight words</p>
22     <p>forty-four words</p>
23   </section>
24 </main>
25 <footer>
26   <p>five words</p>
27 </footer>
```

Semantic Nouns: Native Elements

MDN: [HTML semantic elements grouped functionally](#)

- Content sectioning
 - `<header>`, `<main>`, `<footer>`, `<nav>`
- Text content
 - `<blockquote>`, `<figure>`, `<figcaption>`
- Demarcating edits
 - `<del>`, `<ins>`
- Forms
 - `<button>`, `<fieldset>`, `<legend>`, `<label>`
- Interactive elements
 - `<details>`, `<summary>`

```

1  <!-- Reengineered (functions of a) button -->
2
3  <!-- HTML -->
4  <div id="submit-btn" class="btn" role="button" tabindex="0" aria-disabled="false">Submit</div>
5
6  <script>
7  const btn = document.getElementById('submit-btn');
8
9  // 1. Handle mouse clicks
10 btn.addEventListener('click', handleSubmit);
11
12 // 2. Handle keyboard - Enter key
13 btn.addEventListener('keydown', (e) => {
14 |   if (e.key === 'Enter') handleSubmit();
15 | });
16
17 // 3. Handle keyboard - Space key
18 btn.addEventListener('keyup', (e) => {
19 |   if (e.key === ' ') handleSubmit();
20 | });
21 |
22 // 4. Handle disabled state
23 function disableBtn() {
24 |   btn.setAttribute('aria-disabled', 'true');
25 |   btn.setAttribute('tabindex', '-1');
26 |   btn.removeEventListener('click', handleSubmit);
27 |   // Note: keyboard re-activation still not fully blocked
28 | }
29
30 // 5. Handle form submission
31 function handleSubmit() {
32 |   const form = btn.closest('form');
33 |   if (form) form.requestSubmit();
34 | }
35 </script>
36
37 <!-- CSS - focus ring -->
38 <style>
39 .btn:focus-visible {
40 |   outline: 2px solid #005fcc;
41 |   outline-offset: 2px;
42 | }

```

Our phx-click doesn't make it a Button

Please don't

```
<div class="btn-primary"  
  phx-click="save">Save</div>
```

- ✗ No “button” role in the accessibility tree
- ✗ Not in the tab order
- ✗ Keypress Enter and Space do nothing

Let's do

```
<button type="button"  
  phx-click="save">Save</button>
```

- ✓ Announced as “button” for free
- ✓ Focusable and in the tab order
- ✓ Enter and Space activate it

Let's talk about ARIA: The Golden Rules

- 1 If a native element exists, use it.
- 2 Don't override native semantics unless you absolutely must.
- 3 No ARIA is better than bad ARIA. 

```
<button aria-expanded="false" onclick="toggleMenu()">Menu</button>
```

```
1  def render(assigns) do
2      ~H"""
3      <button aria-expanded={@menu_open} phx-click="toggle_menu">
4          Menu
5      </button>
6      <ul :if={@menu_open} id="menu">
7          ...
8      </ul>
9      """
10 end
11
12 def handle_event("toggle_menu", _params, socket) do
13     {:noreply, assign(socket, menu_open: !socket.assigns.menu_open)}
14 end
15
```

Let's talk some more about ARIA: "Only Use If"

- Content changes without a page reload (e.g., errors/success, chat messages)
 - ✓ use `aria-live="polite"` or `aria-live="assertive"`.
- Multiple landmarks of the same type (e.g., two `<nav>` elements)
 - ✓ add `aria-label="Primary Navigation"`, ...
- Inputs and buttons without visible labels
 - ✓ add e.g. `aria-label="Search"`

In Accessibility: Redundancy is King!

Error Classic Flavoured

Please enter a valid email address.

Error Common Sense Edition

 **Error:** Please enter a valid email address.

If you remove any one: color, icon, text ... the message still lands 👍

How to pass the Code Review: Manual Testing

- Keyboard Navigation:
 - **Tab & Tab+Shift:** Move back & forth between interactive elements.
 - **Enter:** Trigger buttons and links.
 - **Space bar:** Trigger buttons, checkboxes, and scroll.
 - **Arrow keys:** Navigate within menus, radio groups, ...
- Dev Tools:
 - Use the emulate vision deficiencies*).
 - Google Lighthouse: Bad accessibility, Good accessibility

*) CMD+Shift+P → Show Rendering → Emulate vision deficiencies

How to pass the Code Review: Automated Testing

- **CI/CD Pipeline:** axe-core checks run automatically
 - a11y_audit + Wallaby: axe-core checks as ExUnit assertions, right inside `mix test`

MR/PR Checklist

- Is there a native element for this?
- Can I keyboard through my app?
- Does every *non-decorative* icon/image have a text name?
- Does the color contrast pass?
- Did automated checks pass in the CI/CD pipeline?

04

A little Quiz

Question 1

Which of these HTML elements has zero semantic meaning to a screen reader?

1) `<br>`

2) `<p>`

3) `<div>`

4) `<strong>`

Question 2

If a designer gives you a button that navigates the user to a new page, you should use a `<button>` element.

1) FALSE

2) TRUE

Question 3

You have a decorative icon of a magnifying glass next to a labelled search input.

What should the alt attribute look like?

1) `alt="Search"`

2) `alt="Magnifying glass"`

3) `alt=""`

4) Omit the alt attribute entirely

Question 4

`<div phx-click="save">Save</div>`

... is as accessible as ...

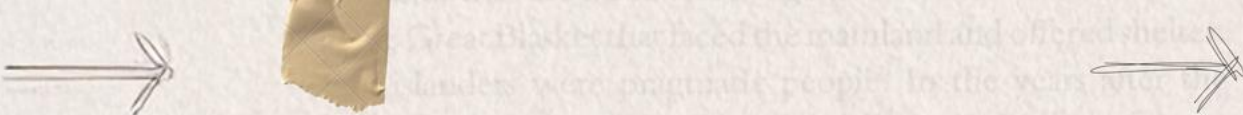
`<button phx-click="save">Save</button>.`

1) True

2) False

05

Future Mindset



2. KAIZEN (改善)

1% Better Every Day

Forget massive transformations.

Japanese companies dominate
through tiny, daily improvements.

Today's micro-win becomes
tomorrow's competitive advantage.



What's your 1% for today?

Thank you

get in touch



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curiosdevcookie.dev



@curiosdevcookie

List of sources of this talk

- WCAG by W3C: <https://www.w3.org/WAI/WCAG21/Understanding/>
- WCAG Testing Checklist: <https://www.accessguide.io/wcag-testing-checklist>
- Native Elements: <https://developer.mozilla.org/en-US/docs/Web/HTML/Reference/Elements>
- Good and bad a11y sites comparison: https://openassessittoolkit.github.io/accessible_u

List of great sites for accessibility

- The A11Y Project:
<https://www.a11yproject.com>
- A11y Cat:
<https://raindrop.io/a11ycat>
- A, AA, AAA:
<https://www.a11yproject.com/checklist/>
- All about screen readers: <https://tetralogical.com/blog/2025/10/14/common-misconceptions-about-screen-readers/>